

HARIHARAN NARLAKANTI

AI Engineer | NLP | LLM Applications | Backend

+91 8686937565 | narlakantihariharan@gmail.com | Hyderabad, India | LinkedIn | GitHub | Portfolio

SUMMARY

Computer Science graduate with **8.89/10 CGPA** focused on AI engineering, NLP, and LLM applications. Hands-on experience building transformer-based systems, RAG pipelines, semantic retrieval, and AI services using **Python, PyTorch, Hugging Face, FastAPI, and vector databases**.

EDUCATION

Malla Reddy University, Hyderabad

2022 – 2026

Bachelor of Technology – Computer Science & Engineering (AI & ML), CGPA: **8.89 / 10**

TECHNICAL SKILLS

AI/ML: Python, PyTorch, Hugging Face Transformers, SentenceTransformers, NLP, RAG, Embeddings, QLoRA, Wav2Vec2

LLM/Retrieval: LLM Applications, ChromaDB, Semantic Search, Vector Retrieval, Model Evaluation

Backend/Tools: FastAPI, REST APIs, Docker, Git, Linux, React, Next.js

EXPERIENCE

Full Stack Development Intern – SmartInternz

May 2025 – Jul 2025

- Built **SocialeX**, a full-stack social networking application on the MERN stack, implementing authenticated REST APIs, JWT-based access control, and MongoDB-backed data models.
- Implemented real-time one-to-one messaging via Socket.IO and enforced server-side authorization across user profiles, posts, and protected resources.

PROJECTS

Saaram – AI-Powered Telugu News Summarization & Speech Platform

GitHub | Live Demo

- Built a hybrid Telugu summarization pipeline combining morphology-aware TF-IDF, mT5 transformers, and **QLoRA** fine-tuning with adaptive routing and resource-aware fallback inference.
- Evaluated **5 summarization configurations on 1,302 XL-Sum Telugu samples**; fine-tuned mT5 achieved **0.1639 ROUGE-1, 0.0496 ROUGE-2, and 0.1452 ROUGE-L**.
- Improved character n-gram TF-IDF **ROUGE-2 by 26.6%** over the word-level baseline and deployed the inference system through FastAPI, Docker, and Hugging Face Spaces.

ResearchCompass – AI Research Paper Analyzer

GitHub | Live Demo

- Engineered a research-paper analysis pipeline using **FastAPI, PyMuPDF, ChromaDB, SentenceTransformers, and Groq**, supporting semantic search, cross-paper comparison, and literature review generation.
- Implemented section-aware chunking, **384-dimensional BGE embeddings**, targeted multi-query retrieval, context deduplication, and structured Pydantic-validated LLM outputs.
- Built **5 REST workflows** and validated ingestion, chunking, vector search, and synthesis through **49 unit/integration tests**.

VoiceVibes AI – Vocal Emotion Analytics Dashboard

GitHub

- Built a speech-emotion analysis system using a fine-tuned **Wav2Vec2 transformer** with CPU inference behind a FastAPI backend.
- Implemented browser-native audio capture and custom mono 16-bit PCM WAV encoding, with probability visualization across **7 emotion classes**.
- Resolved legacy checkpoint compatibility by mapping outdated parameter names and projection dimensions to the current Transformers architecture.

ACHIEVEMENTS & RESEARCH

- Best Project Award**, Malla Reddy University – Telugu NLP news summarization platform.
- Co-authored **Saaram: Resource-Aware Telugu News Summarization with Morphology-Aware TF-IDF and mT5**, submitted to **CIS 2026 (Springer LNNS)**.

CERTIFICATIONS

- Machine Learning Specialization** – DeepLearning.AI & Stanford University (2026)
- IBM Full Stack Software Developer Professional Certificate** – IBM / Coursera (2024)
- AWS Academy Cloud Foundations** – AWS Academy (2025)